Priyanshu Choudhary

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Education

Chandigarh University (CGPA - 8.09) — Gharuan, Punjab
B.E, Computer Science Engineering — *Aug 2023 – Expected May 2027*

Oxford Public School (91%) — Pragati Path Ranchi, Jharkhand
PCM + CS — *March 2021 – April 2022*

Experience

Autonomous Navigation and Computer Vision Intern — May 2025 – June 2025
***Rekhi Foundation**, MIND Lab, Chandigarh University* — *Gharuan, Punjab, India*
Tech: ROS2, SLAM, LiDAR, YOLOv5, RViz, Gazebo, Tesseract, Python

- Built a real-time navigation system with **SLAM** and **LiDAR** on **ROS2** for mapping and localization, optimized for low-latency processing
- Developed a **YOLOv5** object detection pipeline and an **OCR** pipeline using **Tesseract** for text extraction
- Reduced processing overhead by subscribing selectively to only the required **ROS2** topics (e.g., filtered **LiDAR** scans and camera frames) and skipping unused data, cutting latency from **210 ms** to **85 ms** (∼**60%**)

Projects

Campus Connect V2.0 | GitHub
Tech: Node.js, Express.js, MongoDB, Redis, WebSockets, Cloudinary — Oct 2026 – Nov 2026

- Built a full-stack campus social platform with real-time chat, stories, notifications and peer discovery over WebSockets.
- Cut feed latency by **97%** and story latency by **91%** by adding Redis caching with TTL, reducing response times from **450 ms** to **11 ms** (Postman, 1000 req, local testing).
- Designed REST APIs with cursor-based pagination for infinite scroll, Cloudinary media uploads, and MongoDB TTL indexes that auto-expire stories after **24 hours**.
- Secured the platform with bcrypt password hashing, migration of legacy plaintext passwords, Redis-backed sessions, and rate limiting on endpoints (**20 requests** per **15 minute** window).

Intelligent Document Processing System | GitHub
Tech: Go, Kafka, PostgreSQL, MongoDB, MinIO, Apache Tika, ClamAV, Next.js — Sept 2026 – Oct 2026

- Built an event-driven document pipeline in Go with Kafka, decoupling uploads from ClamAV scanning, Tika extraction and indexing, cutting upload response time by **97.5%** (**800 ms** to **20 ms**).
- Processed documents across file types at **25 docs/min** using **3** Kafka consumers, with retries for failed messages.
- Implemented JWT authentication with bcrypt-hashed credentials in PostgreSQL, and per-user MinIO isolation enforced by ownership checks on **5** endpoints, returning 401/403 on unauthorized access.
- Blocked malware uploads by scanning every file with ClamAV before processing, wrote Go unit tests for auth middleware and cross-user access control.

Technical Skills

Languages: C++, Go, Python, JavaScript, Bash

Core CS: Data Structures & Algorithms, OOP, System Design, DBMS, Computer Networks, Operating System

Development: Node.js, Express.js, Go (Echo), Kafka, WebSockets, JWT, REST APIs, Next.js, React

Databases: MongoDB, PostgreSQL, Redis

Tools: Git, GitHub, Postman, Docker, Linux

Robotics & AI: ROS2, SLAM, LiDAR, YOLOv5, Pandas, NumPy

Achievements

- **Codeforces**: 1601 (Expert), **LeetCode**: 500+ problems solved
- NPTEL Cloud Computing Certification (Elite), IIT Kharagpur – Top 5%, 83% score
- Secretary, ROBO-VERSE Club (Chandigarh University), leading robotics and AI technical initiatives
- Contributing Author, Book Chapter on Large Language Models (LLMs), Elsevier (Under Publication)
- Co-Author, research paper on blockchain technology and decentralized systems, I3CSET-2025 (Under Publication)